

FSPP0034 PHYSICS II
Chapter 1

Tutorial 1

1. What is the magnitude of the force a $+25\ \mu\text{C}$ charge exerts on a $2.5\ \text{mC}$ charge $28\ \text{cm}$ away?
[Ans: $7200\ \text{N}$]
2. Particles of charge $+65$, $+48$ and $-95\ \mu\text{C}$ are placed in a line (Fig.1). The center one is $0.35\ \text{m}$ from each of the others. Calculate the net force on each charge due to the other two.

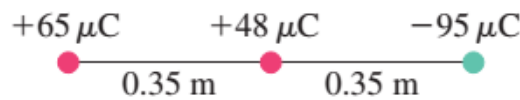


Figure 1

$F_{\text{left}} = 120\ \text{N}$, to the left;
 $F_{\text{center}} = 560\ \text{N}$, to the right;
 $F_{\text{right}} = 450\ \text{N}$, to the left.]

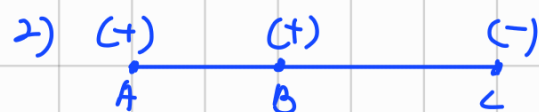
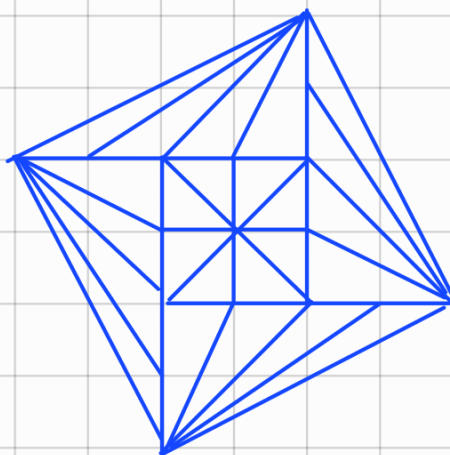
3. What are the magnitude and direction of the electric force on an electron in a uniform electric field of strength $1920\ \text{N/C}$ that points due east ?
[Ans: $3.08 \times 10^{-16}\ \text{N}$ west]
4. Determine the magnitude and direction of the electric field $16.4\ \text{cm}$ directly above an isolated $33.0 \times 10^{-6}\ \text{C}$ charge.
[Ans: $1.10 \times 10^7\ \text{N/C}$ up]
5. Two point charges, $Q_1 = -25\ \mu\text{C}$ and $Q_2 = +45\ \mu\text{C}$ are separated by a distance of $12\ \text{cm}$. The electric field at the point P (see Fig. 2) is zero. How far from is P?
[Ans: $35\ \text{cm}$]



Figure 2

6. The field just outside a 3.50-cm -radius metal ball is $6.25 \times 10^2\ \text{N/C}$ and points toward the ball. What charge resides on the ball?
[Ans: $-8.52 \times 10^{-11}\ \text{C}$]

$$\begin{aligned}
 1) \quad F &= \frac{k |q_1| |q_2|}{r^2} \\
 &= \frac{9 \times 10^9 \times 25 \times 10^{-6} \times 2.5 \times 10^{-3}}{0.28^2} \\
 &= 7174.74 \text{ N}
 \end{aligned}$$



$$\begin{aligned}
 \Sigma F_A &= F_{AC} - F_{AB} \\
 &= \frac{9 \times 10^9 \times 95 \times 10^{-6} \times 65 \times 10^{-6}}{0.7^2} - \frac{9 \times 10^9 \times 48 \times 10^{-6} \times 65 \times 10^{-6}}{0.35^2} \\
 &= 113.42 - 229.22 \\
 &= -115.8 \text{ N (to the left)}
 \end{aligned}$$

$$\begin{aligned}
 \Sigma F_B &= F_{BC} + F_{BA} \\
 &= \frac{9 \times 10^9 \times 48 \times 10^{-6} \times 65 \times 10^{-6}}{0.35^2} + \frac{9 \times 10^9 \times 48 \times 10^{-6} \times 95 \times 10^{-6}}{0.35^2} \\
 &= 229.22 + 335.02 \\
 &= 564.24 \text{ N (to the right)}
 \end{aligned}$$

$$\begin{aligned}
 \Sigma F_C &= -F_{CB} + (-F_{CA}) \\
 &= -\frac{9 \times 10^9 \times 65 \times 10^{-6} \times 95 \times 10^{-6}}{0.35^2} + \left(-\frac{9 \times 10^9 \times 48 \times 10^{-6} \times 95 \times 10^{-6}}{0.7^2} \right) \\
 &= -113.42 - 335.04 \\
 &= -448.47 \text{ N (to the left)}
 \end{aligned}$$

$$3) \quad E = +1920 \text{ N/C}, \quad q = 1.6 \times 10^{-19}$$

$$F = \frac{F}{q}$$

$$F = E_q = 1920(-1.602 \times 10^{-19}) = 3.0758 \times 10^{-16} \text{ to the west (-)}$$

$$4) r = 16.4 \text{ cm}, q = 33.0 \times 10^{-6} \text{ C}$$

$$E = \frac{kQ}{r^2} \\ = \frac{9 \times 10^9 (33 \times 10^{-6})}{0.164^2} \\ = 1.104 \times 10^6 \text{ N/C} \text{ upward}$$

$$5) Q_1 = -25 \times 10^{-6}, Q_2 = +45 \times 10^{-6}, r = 12 \text{ cm}$$

$$E_{P1} = E_{P2}$$

$$\frac{Q}{r^2} = \frac{Q}{r^2}$$

$$\frac{25 \times 10^{-6}}{x^2} = \frac{45 \times 10^{-6}}{(x+12)^2}$$

$$x = -5.1, x = 35.1 \text{ cm} \\ (\text{not accepted})$$

$$6) E = 6.25 \times 10^2 \text{ N/C}, r = 3.5 \text{ cm}$$

$$E = \frac{kQ}{r^2}$$

$$6.25 \times 10^2 = \frac{9 \times 10^9 (Q)}{(3.5 \times 10^{-2})^2}$$

$$Q = 8.507 \times 10^{-11} \text{ C}$$

$$8) q = -1.602 \times 10^{-19}, v_i = 5 \times 10^5 \text{ m/s}, v_f = 0$$

$$V = \frac{U}{q_0}$$

$$= \frac{\frac{1}{2} (9.1 \times 10^{-31}) (5 \times 10^5)^2}{-1.602 \times 10^{-19}}$$

$$= -0.71 \text{ V}$$

$$9) r = 0.5 \times 10^{-10} \text{ m}, V = 0, r = \infty$$

$$Q = +e = 1.60 \times 10^{-19} \text{ C}$$

$$V = \frac{Q}{4\pi\epsilon_0 r} = \frac{1.60 \times 10^{-19}}{4 \times \pi \times 8.85 \times 10^{-12} \times 0.5 \times 10^{-10}} = 29 \text{ V}$$

$$(b) Q = -e = -1.60 \times 10^{-19} \text{ C}$$

$$U = qV = -1.60 \times 10^{-19} (29 \text{ V})$$

$$= -4.64 \times 10^{-18} \text{ J}$$

8. What potential difference is needed to stop an electron that has an initial velocity, $v = 5 \times 10^5$ m/s. [Ans: -0.71 V]

9. (a) What is the electric potential 0.50×10^{-10} m from a proton? Let $V = 0$ at $r = \infty$.

(b) What is the potential energy of an electron at this point?

[Ans: (a) 29 V (b) -4.6×10^{-18} J]

10. Point a is 26 cm north of a $-3.8 \mu\text{C}$ point charge, and point b is 36 cm west of the charge (Fig. 3). Determine (a) $V_b - V_a$ and (b) $\vec{E}_b - \vec{E}_a$ (magnitude and direction).

[Ans: (a) 3.6×10^4 V (b) 5.7×10^5 V/m, 62°]

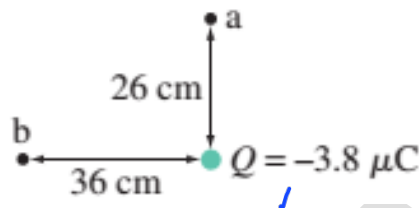


Figure 3

$$10) (a) V_a = \frac{kQ}{r} = \frac{9 \times 10^9 (-3.8 \times 10^{-6})}{0.26}$$

$$= -1.316 \times 10^5$$

$$V_b = \frac{kQ}{r} = \frac{9 \times 10^9 (-3.8 \times 10^{-6})}{0.36}$$

$$= -9.5 \times 10^4$$

$$V_b - V_a = -9.5 \times 10^4 - (-1.316 \times 10^5)$$

$$= 36600 \text{ V}$$

$$(b) E_a = \frac{kQ}{r^2} = \frac{9 \times 10^9 (3.8 \times 10^{-6})}{0.26^2}$$

$$= 505917 \text{ N/C}$$

$$E_b = \frac{kQ}{r^2} = \frac{9 \times 10^9 (3.8 \times 10^{-6})}{0.36^2}$$

$$= 263888 \text{ N/C}$$

	x	y
E_a	0	505917
E_b	263888	0
ΣE	263888	505917

$$\sqrt{263888^2 + 505917^2}$$

$$= 570604 \text{ N/C}$$

$$\theta = \tan^{-1} \left(\frac{505917}{263888} \right)$$

$$= 62.45^\circ$$